

29	Electron diffraction from a crystal occurs when the wavelength of the electrons is comparable to the atomic spacing of the atoms in the crystal. Atomic spacing is about 10^{-10} m. Free electrons are accelerated from rest by a potential difference V in a vacuum before being incident onto the crystal. What would be a suitable value for V?							
	A	1 mV	B	1 V	C	1 kV	D	1 MV

Answer: C

$$\lambda = \frac{h}{p}, E_K = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mE_K}, E_K = eV$$

$$\text{so, } \lambda = \frac{h}{\sqrt{2meV}} \Rightarrow V = \frac{1}{2me} \left(\frac{h}{\lambda} \right)^2 = \frac{1}{2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19}} \left(\frac{6.63 \times 10^{-34}}{10^{-10}} \right)^2 = 150 \text{ V}$$

30 The deviation of α particles by thin metal foils through angles that range from 0° to 180°